Valupadasu Revanth

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RESEARCH INTERESTS

Explainable AI (XAI), Trustworthy AI, Reinforcement Learning, Computer Vision, Cyber-Physical Systems, Edge AI/TinyML, Human-AI Interaction, Sensor-Driven Intelligent Systems.

EDUCATION

Indian Institute of Technology | Kharagpur — Jul 2025 - Apr 2027
M.Tech in Cyber Physical Systems — *CGPA : 8.23/10.0*

Vaagdevi College of Engineering | Warangal — Dec 2021 - Apr 2025
B.Tech in Computer Science and Engineering — *CGPA : 8.49/10.0*

RESEARCH EXPERIENCE & PROJECTS

Explainable and Secure Reinforcement Learning Framework for Smart Automobiles | View Report — Apr 2026 - Present

M.Tech Major Project under Dr. Ayantika Chatterjee

- Designed a modular dual-verification explainability system for deep reinforcement learning in CARLA, targeting trustworthy decision-making on resource-constrained edge platforms. Developed a **semantic-mask-gated Grad-CAM method** using an auxiliary linear probe to disentangle VAE latent representations and generate localized visual explanations. Created a gradient-free decoder reconstruction-difference method to independently verify latent-representation fidelity, achieving **0.90 AUPRC** and **1.00 recall** for critical dynamic objects under severe class imbalance.
- Research Methods: Explainable AI, Representation Learning, Reinforcement Learning, Class-Imbalance Analysis, Latent-Space Analysis, Visual Explanation.

Technologies / Tools Used : *CARLA, SHAP, Linear Probing, Python, PyTorch, TensorFlow, PPO, VAE, Grad-CAM*

ML Equivalent Control Strategy for Dual-Aware ACC | View Report Github — Jan 2026 - Apr 2026

CPS Application Lab Project under Dr. Somnath Sengupta

- Engineered a miniature **Adaptive Cruise Control** testbed on a **Raspberry Pi 4**, integrating dual ultrasonic sensors and wheel encoder feedback for robust real-time environmental awareness; designed a cascaded dual-aware **Model Predictive Control** architecture utilizing Hildreth's Quadratic Programming to proactively adjust speeds based on front and rear vehicle proximities.
- Executed a complete **TensorFlow-to-NumPy migration** of an MLP surrogate model, achieving **sub-0.4 ms inference latency** to bypass heavy optimization and enable seamless edge deployment; validated the dual-aware MLP controller on physical hardware, correctly triggering speed adjustments during rear approaches and reducing velocity-tracking RMSE by **50%** with $\mathbf{R^2}$ **exceeding 0.98**.

Technologies / Tools Used : *Python, TensorFlow, Raspberry-Pi, Flask, NumPy, MPC, PID, MLP*

Human Activity Recognition with K-Means Feature Selection | Github — Nov 2025 - Dec 2025

Independent Project

- Implemented K-Means-based unsupervised feature selection on **561 smartphone sensor features**, improving Gaussian Naive Bayes activity-recognition accuracy from **73.15% to 78.59%** (**+5.44 percentage points**).
- Reduced the feature set from **561 to 50 representative features** (**91% reduction**), achieving approximately **92% lower Gaussian Naive Bayes execution time** while improving classification accuracy; evaluated the accuracy–efficiency trade-off across **10–100 selected features**, measuring classification accuracy, model execution time, and K-Means feature-selection overhead for sensor-driven healthcare analytics.

Technologies / Tools Used : *Python, Scikit-learn, K-Means Clustering, Gaussian Naive Bayes, Pandas, NumPy, Matplotlib, UCI HAR Dataset*

TECHNICAL SKILLS

AI/ML :	PyTorch, TensorFlow, Scikit-learn, PPO, VAE, Grad-CAM, SHAPBPT, Linear Probing, OpenCV
Research Areas :	Explainable AI, Trustworthy AI, Reinforcement Learning, Computer Vision, Cyber-Physical Systems
Programming :	Python, C++, C, Embedded C
Embedded & Protocols :	Raspberry Pi, ESP32, ATmega, UART, SPI, I2C, CAN
Tools :	CARLA, Linux, Git

AWARDS & ACHIEVEMENTS

- Ranked in the **top 1.6%** for GATE 2025 in Computer Science, highlighting strong theoretical and applied computer science expertise.
- Ranked in the **top 5%** for GATE 2025 in Data Science & AI, showcasing highly competitive analytical and problem-solving skills.
- MoE GATE Fellowship (Govt. of India): Receiving **₹12,400/month** for M.Tech studies at IIT Kharagpur based on GATE performance.
- Undergraduate Merit Scholarships: Awarded **₹60,000 annually** for four years, funded jointly by the Government of Telangana (**₹35,000/yr**) and Vaagdevi College of Engineering (**₹25,000/yr**) for continuous academic excellence.

TEACHING EXPERIENCE

Teaching Assistant — Jul 2026 - Present

Indian Institute of Technology Kharagpur

- **CPS Architectures: Design and Validation:** Delivered lecture on Embedded C essentials, educating students on memory architecture, pointer-based direct hardware access, bitwise manipulation, and interrupt handling (ISRs) for resource-constrained environments.
- **Embedded Control Laboratory:** Instructed students on ATmega32 microcontroller configuration and assembly-level debugging using Microchip Studio and Proteus. Guided hands-on hardware interfacing of LCDs and 7-segment displays using Assembly and Embedded C.
- **Embedded Machine Learning:** Facilitated edge computing projects by distributing Arduino BLE boards and mentoring students on deploying and executing machine learning models directly onto physical microcontrollers.